Determinant of 3x3 matrices

Key Points

Finding the determinant of a 3x3 matrix requires decomposition into three 2x2 determinants of minor matrices

1. Choose a row/column to expand along
2. Use the alternating sign matrix to get the multipliers
3. Find the minors and compute the determinants with associated multipliers
4. Evaluate sum

Basic Skills

Q1. Find the following 2x2 determinants

a) $\begin{vmatrix} -3 & 2 \\ -5 & 1 \end{vmatrix}$ b) $\begin{vmatrix} 2 & 4 \\ -1 & -2 \end{vmatrix}$ c) $\begin{vmatrix} -5 & 2 \\ 3 & 0.5 \end{vmatrix}$

Q2. For the matrix $M = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 5 \\ 0 & 3 & -2 \end{pmatrix}$

a) State the minor matrix for element m_{13} (row 1, column 3)

b) State the minor matrix for element m_{22} (row 2, column 2)

Q3. State the alternating sign matrix

Q4. Evaluate $\begin{vmatrix} 2 & -1 & 0 \\ 0 & -2 & 3 \\ 5 & 4 & 2 \end{vmatrix} = 2\begin{vmatrix} -2 & 3 \\ 4 & 2 \end{vmatrix} - -1\begin{vmatrix} 0 & 3 \\ 5 & 2 \end{vmatrix}$

Competency Skills

Q5. For the matrix $M = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 5 \\ 2 & 0 & -2 \end{pmatrix}$ explain why $|M| = 3\begin{vmatrix} 2 & 5 \\ 0 & -2 \end{vmatrix} - \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} + 2\begin{vmatrix} 2 & 1 \\ 2 & 5 \end{vmatrix}$ and hence find $|M|$

Q6. Find the determinant for the following matrices

a) $\begin{pmatrix} 5 & 3 & -2 \\ 2 & 0 & 4 \\ -1 & 1 & 2 \end{pmatrix}$ b) $\begin{pmatrix} 6 & -4 & 1 \\ 0 & 2 & 3 \\ 5 & -3 & 5 \end{pmatrix}$ c) $\begin{pmatrix} -2 & -3 & 4 \\ 8 & -7 & -2 \\ 10 & 3 & -3 \end{pmatrix}$

Q7. For $\begin{pmatrix} \lambda & 1 & 3 \\ -1 & \lambda & 2 \\ 4 & 3 & \lambda \end{pmatrix}$ when the determinant is 3 $\lambda = -4$ is a solution find the others

Advanced Skills

Q8. Explain the geometric interpretation of the determinant of a 3x3 matrix

Q9. Given that the determinant of $\begin{pmatrix} x & 1 & 1 \\ 0 & -x & 3 \\ 2x & 2 & 1 \end{pmatrix}$ is greater than 4 determine the possible values of x

Q10. Deduce how the following operations on a 3x3 matrix affect the determinant

a) Multiplying a single row by a constant

b) Exchanging two rows

c) Scalar multiplication by a constant

Extension

Q11. Given $\begin{vmatrix} 1 & 8 & 10 \\ 2 & 4 & 2 \\ 4 & -2 & -1 \end{vmatrix} = -120$ find $\begin{vmatrix} x^2 & -4 & 10y \\ 2x^2 & -2 & 2y \\ 4x^2 & 1 & -y \end{vmatrix}$

Hint: Consider what operations have happened to each of the columns